ENEE 323 Project 4

Hilbert Transform and Single-Sideband Modulation

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Part (i): FIR Coefficients

Problem i

Plot the coefficients of the given FIR filter against $0 : 2M$, identify the symmetry and structure, and compare the result to the ideal discrete-time Hilbert transformer impulse response.

Solution i

The coefficient vector has length

$$2M + 1 = 81, \qquad M = 40.$$

The plot shows the standard structure of a finite-length Hilbert transformer approximation:

- the sequence is antisymmetric about the center index $n = M$,
- the center coefficient is exactly zero,
- the nonzero terms occur mainly on odd offsets from the center,
- the values decay in a way that follows the ideal Hilbert transformer trend.

The ideal discrete-time Hilbert transformer is

$$h_{\text{ideal}}[n] = \begin{cases} \dfrac{1-(-1)^n}{\pi n}, & n \neq 0, \\ 0, & n = 0. \end{cases}$$

Since

$$1-(-1)^n = \begin{cases} 0, & n \text{ even}, \\ 2, & n \text{ odd}, \end{cases}$$

the ideal response is zero on even indices and behaves like

$$\frac{2}{\pi n}$$

on odd indices. The given FIR filter is therefore a shifted, truncated approximation of the ideal response centered at $n = M$.

From the script output:

$$M = 40, \qquad \text{center coefficient} = 0, \qquad \text{max odd-symmetry error} = 0.$$

This confirms exact odd symmetry in the supplied coefficient vector.

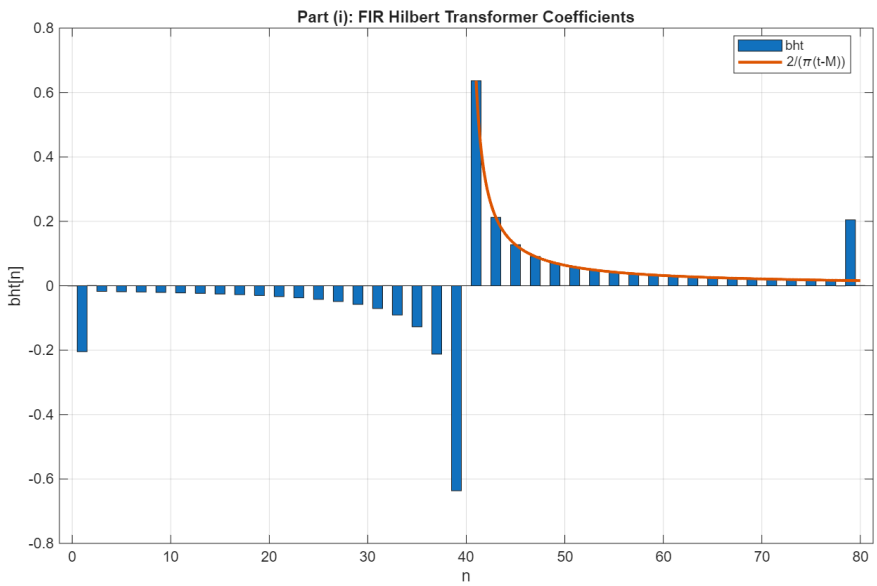

Figure 1. Coefficient sequence of the FIR Hilbert transformer approximation.

Part (ii): Frequency Response

Problem ii

Use a zero-padded DFT with 3200 frequency samples to compute the complex frequency response. Plot the magnitude in dB, identify the passband and ripple, and examine

the phase after unwrapping and normalizing by π.

Solution ii

The frequency response was obtained using a 3200-point FFT of the coefficient vector. The magnitude response shows the expected Hilbert-transformer behavior: the usable passband lies away from both DC and Nyquist, while the edges show stronger deviation from the ideal flat response.
For measurement, the passband used in the script was

$$[0.05\pi,\ 0.95\pi] \text{ rad/sample.}$$

Using the sampling rate from the project, this corresponds to

$$[800,\ 15200] \text{ Hz.}$$

The measured peak-to-valley ripple in that band is

$$7.2198 \text{ dB.}$$

The phase response is close to linear over the passband. This is what we expect from an antisymmetric FIR filter with linear phase behavior and group delay determined by the center index. The slope is tied to the delay, while the intercept reflects the $\pm\pi/2$ phase shift associated with a Hilbert transformer.

From the script output:

$$\text{passband} = [800,\ 15200] \text{ Hz}, \qquad \text{ripple} = 7.2198 \text{ dB},$$

$$\text{phase slope (normalized by } \pi) = -12.7324, \qquad \text{intercept at } \omega = 0 = -0.5.$$

The intercept value -0.5 means the phase starts near

$$-\frac{\pi}{2},$$

which is exactly the expected Hilbert-transformer phase shift.

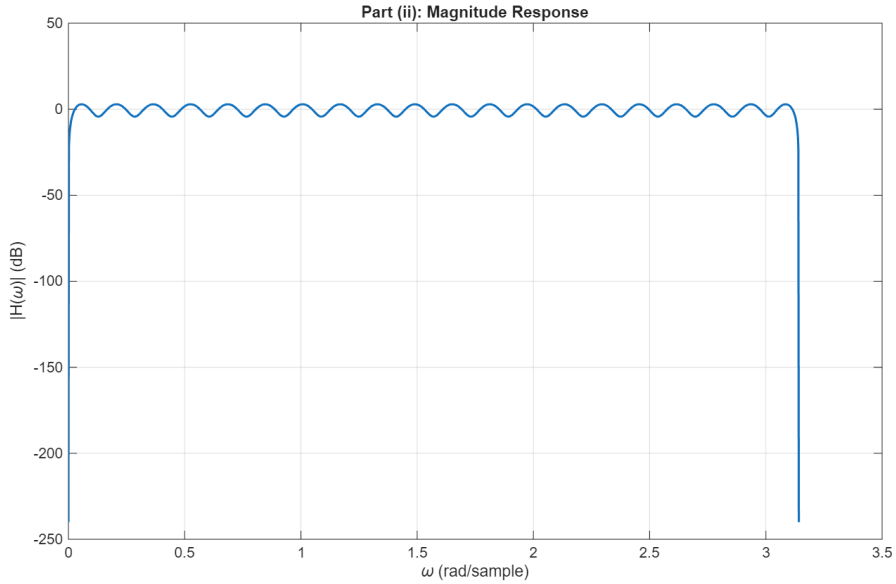

Figure 2. Magnitude response of the FIR Hilbert transformer.

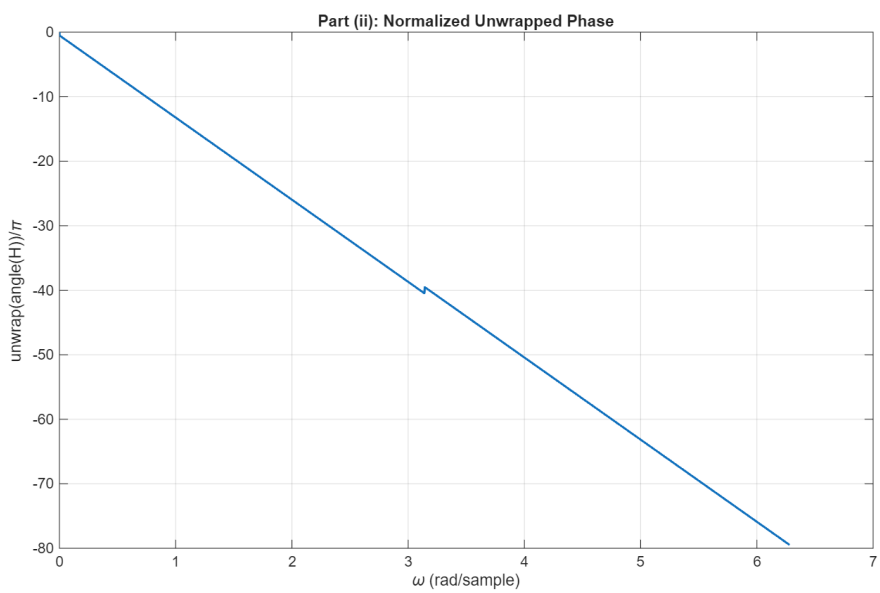

Figure 3. Unwrapped and normalized phase response.

Part (iii): Audio Spectrum

Problem iii

Analyze the signal in `roar.wav`, compute its DFT, determine the frequency band within 60 dB of the maximum magnitude, and compare that band to the filter passband from part (ii).

Solution iii

The audio file was loaded and its DFT magnitude was plotted in dB. The effective occupied band was estimated by locating all frequency bins whose magnitude was within 60 dB of the peak value.
The script reported the band

$$[0,\ 55000]$$

in frequency-index form, and

$$[0,\ 5500]\ \text{Hz}$$

in physical frequency.
Comparing this with the passband from part (ii),

$$[800,\ 15200]\ \text{Hz},$$

the signal band is *not* entirely contained inside the filter passband, because the audio includes content below 800 Hz.

The useful audio content extends down to 0 Hz in the measured thresholded spectrum, while the Hilbert transformer was only measured as reliable from about 800 Hz upward. So the low-frequency portion of the audio is outside the strongest operating region of the filter.

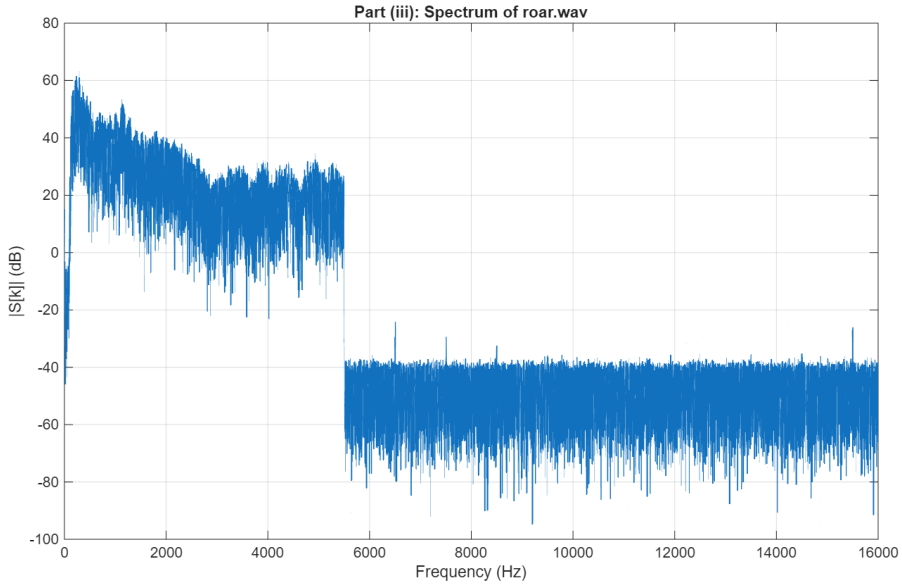

Figure 4. Magnitude spectrum of the audio signal `roar.wav`.

Part (iv): Single-Sideband Modulation

Problem iv

Implement the single-sideband modulation system using the supplied Hilbert transformer approximation and a carrier frequency of $f_c = 6.0$ kHz. Verify that the spectrum is concentrated on one side of the carrier and compute the fraction of energy in the suppressed sideband.

Solution iv

The Hilbert-transform output $\hat{s}[n]$ was computed using convolution with the FIR filter, followed by trimming to account for the delay so that the result matched the original signal length.
The single-sideband signal was formed as

$$x[n] = s[n]\cos(\omega_c n) - \hat{s}[n]\sin(\omega_c n),$$

where

$$\omega_c = 2\pi \frac{f_c}{F_s}, \qquad f_c = 6000 \text{ Hz}.$$

From the spectrum, the energy is concentrated primarily in the upper sideband, while the lower sideband is suppressed. The script measured the missing-sideband energy

fraction as

$$0.009072.$$

This means the unwanted sideband contains less than 1% of the total energy, so the suppression is fairly strong even though the Hilbert transformer is only an FIR approximation.

Script output:

$$\text{kept sideband} = \text{upper sideband}, \qquad \text{missing sideband} = \text{lower sideband},$$

$$\text{fraction of total energy in missing sideband} = 0.009072.$$

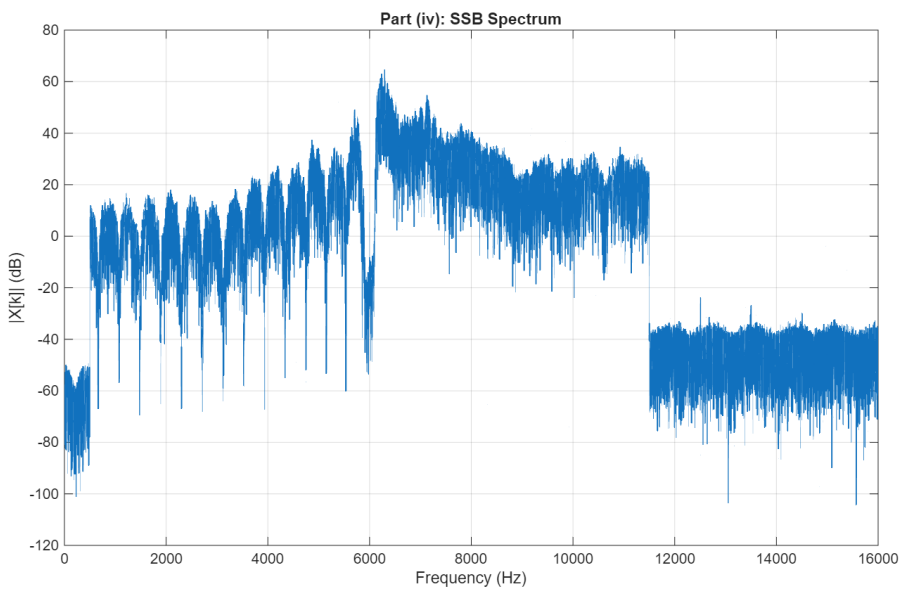

Figure 5. Spectrum of the single-sideband modulated signal.

Part (v): Demodulated Signal

Problem v

Shift the kept sideband back to baseband and form a demodulated version of the original input. Compare the demodulated output to the original signal.

Solution v

After shifting the sideband back down to baseband, the demodulated signal $s_{\text{demod}}[n]$ was compared against the original signal $s[n]$.
The reported relative error is

$$0.50407.$$

So the recovered signal is not identical to the original. This is consistent with the fact that the modulation and demodulation used a practical finite-length Hilbert transformer rather than an ideal one, and the original signal also contains some low-frequency content outside the strongest passband of the filter.

The demodulated signal still resembles the original in structure, but the reconstruction is not especially accurate numerically:

$$\frac{\|s - s_{\text{demod}}\|}{\|s\|} = 0.50407.$$

So the recovered audio is expected to sound similar in some broad features, but noticeably different from the original.

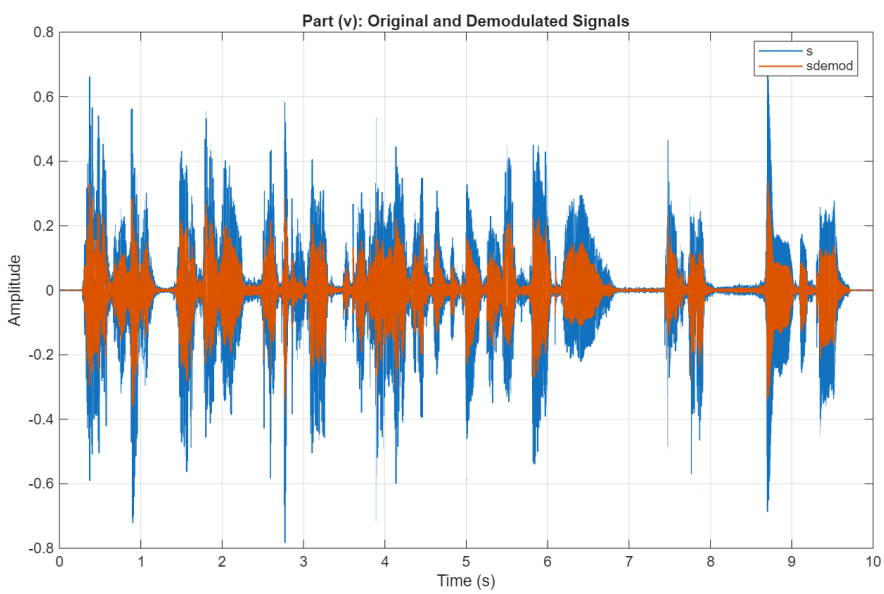

Figure 6. Comparison of the original and demodulated signals.